SE 216 – SOFTWARE PROJECT MANAGEMENT
REQUIREMENTS DOCUMENT

PROJECT NAME: REPLATE

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REQ. #	FUNCTIONAL REQUIREMENTS
1	The system must have separate log in forms for municipalities, business owners and clients.
2	Users must be able to register and log in to the application using their email address or phone number, and create a personal profile.
3	The application must display surplus food products on an interactive map, showing the user's location and nearby branches with available products.
4	Users should be able to filter products by category, price range, distance, expiration date, and stock quantity both on the map and list views.
5	For each listed product, the system should display product name, category, original price, discounted price, expiration date, available quantity, branch name, and pickup time window.
6	Users should be able to reserve a product through the application, and the system must generate a unique reservation code and QR code for pickup verification.
7	Once a reservation is made, the reserved product quantity should be deducted from the available stock and the listing should be updated in real time.
8	Users must be able to cancel their reservation up to 30 minutes before the pickup time, and the cancelled quantity should be made available again for other users.
9	The application should send push notifications to users about new surplus food at nearby branches, products approaching their expiration date, price reductions, and donation campaigns.

10	The system should track and display each user's personal impact dashboard showing total food saved (kg), total money saved (TL), and estimated CO2 emission reduction.
11	Business account management: 11.1 - Businesses must be able to register on the platform. 11.2- The system must verify business accounts before activation. 11.3- Businesses must be able to declare surplus food (name, category, quantity, expiration date, photo, pickup time). 11.4- The system must automatically detect products close to expiration and suggest listing them.
12	Two-stage redistribution model: 12.1- The system must allow users to reserve and collect food directly from businesses within a defined time window. 12.2- If a product is not collected within the defined time window, the system must automatically transfer it to municipal distribution centers.
13	Municipal distribution centers shall be able to view incoming product transfers from businesses, track their current stock, and process user pickups by scanning the reservation QR code.
14	The system should allow businesses to view a dashboard with statistics on their surplus food declarations, waste reduction rates, and donation contributions.
15	The system should automatically detect products within 24 hours of their expiration date and apply dynamic pricing suggestions to encourage faster sales.
16	Users should be able to add branches to their favorites list and receive priority notifications when those branches declare new surplus products.
17	Donation features: 17.1- The system must allow users to donate meals for people in need. 17.2- The system must allow users to donate pet food for animal shelters.

18	Municipal management features: 18.1- The system must allow municipalities to monitor reported surplus food. 18.2- The system must allow municipalities to track transfers to distribution centers. 18.3- The system must allow municipalities to manage redistribution operations. 18.4- The system must allow municipalities to analyze regional food data.
19	The system must support in-app payment processing for discounted surplus food purchases using secure payment methods.
20	The system must automatically update the status of products (e.g., available, reserved, picked-up, expired, donated, bio-waste) based on user actions and time limits.
REQ. #	**NON-FUNCTIONAL REQUIREMENTS**
1	The system must return product search and filter results in less than 3 seconds under normal network conditions.
2	The system must maintain at least 99% uptime availability and be accessible 24/7.
3	The system must be able to restart itself within 5 minutes in case of a crash or unexpected failure.
4	All user personal data, payment information, and reservation data must be stored securely using encryption (AES-256) and transmitted over secure protocols (SSL/TLS).
5	Only authorized administrators should have access to system management panels and sensitive business data.
6	At least 90% of users must be able to complete a product reservation within 3 minutes without external assistance.
7	The system must support at least 10,000 concurrent users with response times below 3 seconds for core operations (search, reserve, update).

8	The system must perform automated data backups every day at 02:00 AM, complete each backup before 03:00 AM, and retain backup data for at least 7 days.
9	The application must support Android 10 and above and iOS 14 and above, and must function correctly on devices with screen sizes between 4.7 inches and 6.7 inches and at least 4GB RAM.
10	The system must be designed using a modular architecture, and new features must be integrated by modifying no more than 15% of the existing non-UI codebase.
11	The map and location-based features must accurately detect the user's GPS location with a maximum error margin of 25 meters.
12	The system must maintain an error rate below 1% during normal operation, apply critical security patches within 48 hours of detection, and ensure that system downtime during updates does not exceed 5 minutes.